

The integrated explicit analytic number theory network

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Disclaimer

- Due to recent events, I have not been able to devote as much time to these slides as I would have liked.

- The traditional unit of mathematical research output is the **paper**.
- Mathematical literature as a whole consists of many papers (and monographs, etc.) linked together.
- But aside from some bibliographic databases (MathSciNet, ZbMath, etc.) or mathematical databases (OEIS, LMFDB, etc.) there are few systematic efforts to organize the literature at larger scales than an individual paper.

- The **Integrated Explicit Analytic Number Theory Network** is an ongoing experiment to see if advances in autoformalization can create a useful new type of mathematical artefact at the scale of multiple papers, rather than a single paper.
- The vision is to combine a large number of papers in this area into a single living “spreadsheet”, formalized in Lean.

What is explicit analytic number theory?

- A core part of **analytic number theory** is concerned with asymptotic bounds involving quantities relating to the prime numbers.
- For instance, one form of the prime number theorem with classical error term that

$$\theta(x) = x + O(x \exp(-c\sqrt{\log x}))$$

for some $c > 0$, where $\theta(x) := \sum_{p \leq x} \log p$ is the Chebyshev theta function.

- The implied constants in the asymptotic notation are usually left **unspecified**, as they are quite tedious to compute.

Generally speaking, these sorts of asymptotic bounds can be satisfactory when analyzing sufficiently large numbers, so long as one does not care to make the threshold for “sufficiently large” explicit. A typical application is

Vinogradov's theorem

Every sufficiently large odd number can be expressed as the sum of three primes.

But when one wants to use analytic number theory to prove a result for *all* numbers, not just the sufficiently large ones, then one needs to make the implied constants in the asymptotic bounds *explicit*, combined with numerical verification to handle small or medium cases. For instance, this was used to prove the non-asymptotic form of Vinogradov's theorem:

Odd Goldbach conjecture (Helfgott 2013)

Every odd number greater than 5 can be expressed as the sum of three primes.

Here is a typical result in explicit analytic number theory:

Explicit prime number theorem (Fiori-Kadiri-Swidinsky 2022)

One has

$$|\theta(x) - x| \leq 121.0961 \left(\frac{\log x}{R} \right)^{3/2} \exp(-2\sqrt{\log x/R})$$

for all $x \geq 2$, where $R = 5.5666305$.

In practice, such results only are useful for extremely large x , e.g., $x \geq e^{100}$, which is too large a threshold to cover remaining ranges by purely numerical methods. So one also seeks “medium-range” estimates, such as the following:

Explicit prime number theorem (Fiori-Kadiri-Swidinsky 2022)

One has

$$|\theta(x) - x| \leq 5.9651 \times 10^{-7} x$$

when $x \geq e^{30}$.

The paper of Fiori-Kadiri-Swidinsky 2022 contains large tables of results of this type.

The explicit analytic number theory literature is highly interdependent. For instance, there is a broad pipeline from results in each of the categories below to the next:

- **Zeta function estimates:** zero-free regions, zero density estimates, etc. for the Riemann zeta function $\zeta(s)$.
- **Primary explicit estimates:** explicit bounds on the Chebyshev function $\psi(x) = \sum_{n \leq x} \Lambda(n)$ or the Mertens function $M(x) = \sum_{n \leq x} \mu(n)$, both of which are tied to the zeta function by various “explicit formulae”.
- **Secondary explicit estimates:** explicit bounds on other expressions relating to primes, such as the second Chebyshev function $\theta(x) = \sum_{p \leq x} \log p$, the prime counting function $\pi(x)$, or bounds on prime gaps.
- **Applications:** explicit results in number theory that rely on the above estimates

A typical zeta function estimate:

Explicit zero-free region (Mossinghoff–Trudgian 2015)

There are no zeroes $\zeta(s)$ of the zeta function with

$$\operatorname{Re}(s) \geq 1 - \frac{R}{\operatorname{Im}(s)}$$

and $R = 5.5666305$.

A typical primary explicit estimate:

Explicit prime number theorem (Fiori-Kadiri-Swidinsky 2022)

One has

$$|\psi(x) - x| \leq 121.096 \left(\frac{\log x}{R} \right)^{3/2} \exp(-2\sqrt{\log x/R})$$

for all $x \geq 2$, where $R = 5.5666305$.

A typical secondary explicit estimate:

Explicit prime number theorem (Fiori-Kadiri-Swidinsky 2022)

One has

$$|\theta(x) - x| \leq 121.0961 \left(\frac{\log x}{R} \right)^{3/2} \exp(-2\sqrt{\log x/R})$$

for all $x \geq 2$, where $R = 5.5666305$.

A typical tertiary explicit estimate. A **highly abundant number** N is a number for which the sum of divisors $\sigma(N)$ is larger than $\sigma(M)$ for all $M < N$.

Is $\text{LCM}(1, \dots, n)$ highly abundant? (MathOverflow, 2025)

The least common multiple of the first n positive integers is highly abundant if and only if $n \leq 70$, $81 \leq n \leq 96$, $125 \leq n \leq 148$, and $169 \leq n \leq 172$.

Secondary explicit estimates were needed to cover the range $n \geq 89693^2$; the remaining cases require computer computation. The entire theorem is verified in Lean (contingent on a tertiary explicit estimate of Dusart (2018)).

- Because of this interdependence, numerical errors in one explicit paper could propagate to other papers, leading to potential trust issues with the literature.
- On the other hand, these results are quite useful for various applications (e.g., Erdős problems).
- This suggests a good use case for **formalization**.
- However, the task of carefully going through the arguments is quite tedious, particularly with regards to the extensive mechanical computations involved.
- This suggests a good use case for **auto-formalization**.

- The literature updates over time due to various numerical or technological improvements, and many of the intermediate results are of no interest to highly curated formal repositories such as Mathlib.
- Furthermore, these formalization tasks are unlikely to compete with existing human-led projects.
- This makes for a opportunity to deploy larger amounts of **AI assistance** than would otherwise be advisable for a mathematical project.

The Integrated Explicit Analytic Number Theory Network

I launched the **Integrated Explicit Analytic Number Theory Network (IEANTN)** in Jan 2026, hosted within the existing **Prime Number Theorem and more (PNT+)** formalization project.

- **Phase 1** of the project seeks to collect a large number of inter-related results in explicit analytic number theory and formalize them all in a single Lean repository.
- **Phase 2** of the project seeks to use AI tools to auto-update the constants in one formalized paper given changes in the constants in the input papers, e.g., to feed in a new zero free region and see how downstream constants change.
- The dream is to obtain a “living spreadsheet” of explicit constants in the subject that can be updated on the scale of hours or days, rather than years, as well as rolled back in time.

Dozens of contributors (including three students under my co-supervision, Xinjie He, Rushil Raghavan, and Hyunsik Chae)



The first six months of the project proceeded like a “classic” formalization effort, with individual lemmas proven by hand. This led to many subprojects completed, including:

- Chebyshev’s theorems $\psi(x), \theta(x) \asymp x, \pi(x) \asymp \frac{x}{\log x}$ (upstreamed to Mathlib)
- Mertens’ theorems (submitted for Mathlib)
- Costa-Pereira’s inequalities (upstreamed to Mathlib)
- Applications of prime gap bounds to Goldbach-type problems
- Ramanujan’s inequality $\pi(x)^2 < \frac{ex}{\log x} \pi\left(\frac{x}{e}\right)$ for sufficiently large x
- Erdős’ 392 conjecture on well-balanced factorizations of $n!$

- More recently, **autoformalization** advanced to the point where most outstanding lemma requests were quickly filled.
- However, the AI-generated proofs were often **extremely** bloated, with many redundancies.
- Gaps in Mathlib (e.g., in contour integration) were replaced by extremely lengthy **ad hoc** patches.
- Building the entire project now took more than an hour, and the code was no longer human readable.
- A major refactor was needed to take into account the new reality of autoformalization.

- A new system was implemented (with heavy AI assistance) in August, but development has stalled due to the broader crisis in mathematics.
- Whereas the original network was based around a blueprint of lemmas, the new IEANTN is based around a blueprint of “nodes”, each of which roughly corresponds to a single published paper.

A node consists of:

- A “Conclusions.lean” file containing the main claims of the node, in a human-readable Lean file.
- A “formalization.yaml” file describing various metadata, for instance what the logical dependencies of each conclusion are, and how the claims are justified: formalized, appeal to literature, computation, conjecture folklore, etc..
- If formalized, a “Solutions.lean” file establishing the main claims of the node (conditional on dependencies). This can be a large AI-generated Lean file, verified by Comparator but not part of the core Lean base.
- Nodes can be created or deleted as needed, as the structure of the mathematical repository evolves.



- I **had** planned to use the month of September to flesh out the network, and get my students to try to build additional tools on top of it.
- This is currently on hold, but I hope to return to the project.

Thanks for listening!